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Education

Franklin University (2024 - Current)

Master of Science in computer science

Courses: Foundations of Programming, Mathematics for Computing, Data Structures and Object-Oriented Design, Introduction to Analytics, Advanced Data Structures and Programming, Analysis of Algorithms, Distributed Systems

Projects

Virtual Machine (JVM) Project

- Built a simplified Java Virtual Machine from scratch supporting arithmetic opcodes, method calls, object creation, field access, and heap memory management.
- Implemented visitor-based code generation and AST traversal.
- Created custom bytecode and a VMThread execution model.

Distributed Microservices Group Project

- Designed and developed Quarkus-based microservices (Customer, Product, Purchase, Report) communicating over gRPC and RabbitMQ.
- Containerized services using Docker, deployed onto Kubernetes clusters using Rancher Desktop.
- Implemented database persistence with PostgreSQL and Hibernate Reactive.
- Built REST and gRPC APIs following best backend development practices.
- Set up RabbitMQ consumers and producers with routing keys and smallrye messaging.

Skills

Programming Languages: Java, Python, SQL,

Backend Frameworks: Quarkus (REST + gRPC microservices), Hibernate ORM (Reactive and JPA),

Messaging Systems: RabbitMQ (SmallRye Reactive Messaging),

Databases: PostgreSQL (Reactive drivers, schema design, querying),

Containerization & Deployment: Docker, Kubernetes (Rancher Desktop, YAML configuration),

Cloud/DevOps Tools: Jib (container image build tool), native image builds for Quarkus,

Version Control: Git, GitHub (branching, rebasing, PRs, Git Graph),

Testing: Unit testing in Java (JUnit), Test-Driven Development (TDD-style for projects),

Other: REST API design, gRPC API design, microservices architecture, reactive programming